Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3481-WE01

Title: MEng Engineering

Power Semiconductor Devices

Time Allowed:	2 hours	
Additional Material provided:	Generic physical constants:- $q = e = 1.60 \times 10^{-19}$ C 1 eV = 1.60×10^{-19} J $m_e = 9.1 \times 10^{-31}$ kg $h = 6.63 \times 10^{-34}$ J s $c = 3.00 \times 10^{8}$ m s^{-1} $\varepsilon_0 = 8.85 \times 10^{-12}$ F $m^{-1} = 8.85 \times 10^{-14}$ F cm^{-1} $k = k_B = 1.38 \times 10^{-23}$ J $K^{-1} = 8.6 \times 10^{-5}$ eV K^{-1} $N_A = 6.02 \times 10^{23}$ $mole^{-1}$	
Materials Permitted:		
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer all questions. Use a separate answer book for each question. All relevant working must be shown. Only authorised calculators. No other material is permitted.

Revision:	

Question 1

A GaSb (gallium antimonide) chip is uniformly doped with acceptors. Donor doping is added to the chip with the following bell-shaped profile $N_D(x) = N_D(0)exp\left(-\frac{x^2}{2s^2}\right)$. The resulting band structure is shown in Fig. Q1.

(1)

(i) Copy the sketch and mark the n-type and p-type segment or segments.

(ii) Determine whether there are net currents in the chip, or if it is in thermal equilibrium. What did you base your conclusion on?

[16%]

(2)

(i) At $x = \pm 1\ \mu m$, the Fermi level is located $200\ meV$ below the intrinsic Fermi level. Assume that, at this position, the donor doping levels are negligible and determine the density of acceptor doping in the chip.

(ii) At $x = 0\ \mu m$, the Fermi level is located $400\ meV$ over the intrinsic Fermi level. Determine the value of $N_D(0)$. For your calculations, you may use $N_A = 10^{15}\ cm^{-3}$ if you did not answer Part 2(i) of this question.

[32%]

(3) Show that the built-in field in a semiconductor in thermal equilibrium is $\mathcal{E} = -\frac{k_B T}{q} \cdot \frac{d \ln n}{dx}$, by demanding zero currents in thermal equilibrium. You may find Eq. (7) useful.

[12%]

(4) Calculate the built-in field across the n-type region in the chip, as a function of location. You may neglect the acceptor contributions in the n-type region. Hence, determine the field magnitude at $x = 0.25\ \mu m$

[20%]

(5) A different semiconductor with a larger band gap has been doped with the exact same profile specified in this question. How would the calculated field change? Draw the band structure diagram for the new semiconductor.

[20%]

continued

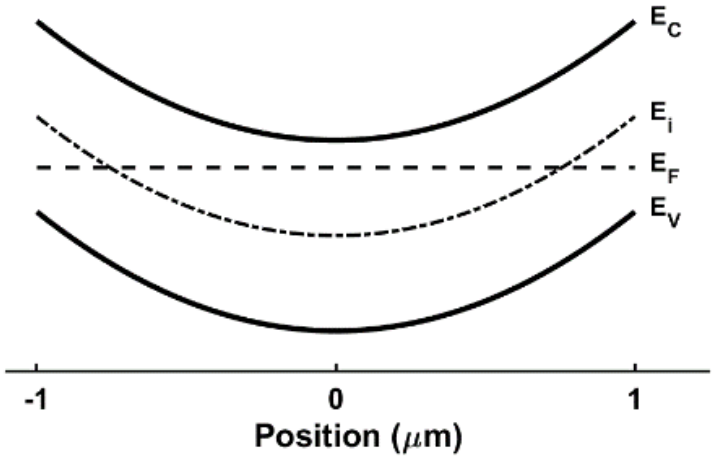


Fig. Q1

Data Q1:

$$s = 0.15\mu m \quad n_i(GaSb) = 1.5 \times 10^{12} cm^{-3}$$

continued

Question 2

A varactor is a device with voltage dependent capacitance, which is used as a tuning element in microwave circuits. Design a varactor based on a uniformly doped p^+-n silicon diode.

(1)

(i) Write an expression for the maximal magnitude of the electric field in the junction, as a function of doping and reverse bias.

(ii) The junction is designed to have a built-in potential of $0.8V$, and operate under reverse bias of $5V$ or less. For the device to function as intended, the maximal magnitude of the electric field must never exceed $30kV/cm$. Find the highest level of donor doping that will comply with these requirements.

[32%]

(2) Find the corresponding acceptor doping level that will ensure that the built-in potential meets the requirements of $V_{bi} = 0.8V$. For your calculations, you may use $N_D = 6 \times 10^{14} cm^{-3}$ if you did not answer part 1(ii) of this question.

[16%]

(3) The varactor is designed to have capacitance variation between $3pF$ and $7pF$. Find the range of reverse bias values that will be required to operate the varactor. For your calculations, you may use $N_D = 6 \times 10^{14} cm^{-3}$ if you did not answer part 1(ii) of this question.

[32%]

(4) It is easier, in some configurations, to realise the varactor using a Schottky junction. Without explicit calculations, explain briefly:

(i) Which side of the junction would you replace with a metal, and which side would you keep as a semiconductor, to keep the device characteristics similar?

(ii) Assuming that we keep the doping level within the semiconductor identical to the corresponding side in the p^+-n junction, which parameter(s) would you need to change to ensure that the capacitance remains the same?

[20%]

continued

Data Q2:

$$n_i = 10^{10} cm^{-3}$$

$$\epsilon_r = 11.7$$

$$A = 10^{-3} cm^2$$

Question 3

3.i. Describe how the maximum operating temperature of a semiconductor device is determined by the bandgap. **[30%]**

3.ii. A schematic cross section of a silicon carbide power diode is shown in figure Q3. Using the parameters listed in table 1, determine the power loss if the diode is operating at a current of 25.0 A and has an area of 1 cm^2. **[20%]**

3.iii. Determine the maximum operating temperature for the diode in figure Q3, using the values in table 1. **[20%]**

3.iv. Power diodes are commonly used in power electronics circuits. An application requires a diode to operate in forward conduction at 10 A and to block 1200 V. The diode switches between the two operating conditions at a frequency of 200 kHz. Identify which type of diode is most suitable for this applicant. Your answer should consider the on-state voltage drop, the reverse leakage current and the switching losses **[30%]**



Figure Q3 : Schematic cross-section of a power diode

Question 4

4.i. Draw a schematic cross-section of a UMOS power Field Effect Transistor. Label the contacts, identify the location of the channel and the depletion region that defines the blocking voltage of a non-punchthrough structure. **[20%]**

4.ii. Calculate the on-state resistance for a MOSFET designed for 1.2 kV operation, if the device is manufactured from

a. Silicon

b. Silicon carbide **[30%]**

4.iii. Determine the on-state resistance at V_{DS}=1 V and the saturation current for a silicon carbide power MOSFET with an oxide thickness of 200 Å, aspect ratio of 1000:1, threshold voltage of 5 V, if the gate voltage is 20 V. **[25%]**

4.iv. Calculate the power dissipated in the MOSFET from part c, if it is operated in a power converter, with a switching frequency of 100 kHz, with a 50% duty cycle. The on-state of the transistor is 10 A at 5 V. The circuit operates at a voltage of 400 V and $\frac{dV}{dt} = 2 \times 10^8$ V/s. **[25%]**

continued

Properties	Si	Ge	GaAs	SiC	Diamond	SiO_2
Dielectric Constant	11.9	16	13.1	9.66	5.5	3.9
Effective Density of States in Conduction Band (cm^{-3})	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}	1.7×10^{19}	1×10^{20}	
Effective Density of States in Valance Band (cm^{-3})	1.04×10^{19}	6.0×10^{19}	7.0×10^{18}	2.5×10^{19}	1×10^{19}	
Effective Mass (Electrons)	$m_l = 0.98$ $m_t = 0.19$	$m_l = 1.64$ $m_t = 0.082$	$m_l = 0.067$	$m_l = 0.29$ $m_t = 0.42$	$m_l = 1.40$ $m_t = 0.36$	
Effective Mass (Holes)	$m_l = 0.16$ $m_t = 0.49$	$m_l = 0.044$ $m_t = 0.28$	$m_l = 0.082$ $m_t = 0.45$	$m_l = 1.00$ $m_t = 1.00$	$m_l = 0.80$ $m_t = 0.80$	
Energy Gap at 300K (eV)	1.12	0.66	1.424	3.23	5.45	9.0
Critical Electric Field (MV/cm)	0.3		0.4	2.5	10	10
Intrinsic Carrier Conc. (cm-3)	1.45×10^{10}	2.4×10^{13}	1.79×10^{6}	10^{-9}	10^{-27}	
Minority Carrier Lifetime (s)	2.5×10^{-3}	10^{-3}	10^{-8}	6×10^{-7}		
Electron Drift Mobility ($cm^2V^{-1}s^{-1}$)	1500	3900	8500	900	1800	
Hole Drift Mobility ($cm^2V^{-1}s^{-1}$)	450	1900	400	400	1500	
Negative Differential Mob. ($cm^2V^{-1}s^{-1}$)				-100		

Table 1 : Properties of Semiconductor Materials at 300 K

END